# ICT COURSE COVERAGE

## COURSE COVERAGE: Class 6

| Sr No | THEME | COMPETENCY | OBJECTIVE | TITLE | TOOLS | Periods | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Demonstration | Hands on & Assessment | Total |
| 1 | Introduction to ICT Environment | Students will be able to develop Collaborative learning, critical thinking & creativity through group activities | ▪ To learn about the ICT environment, Computer systems, and computer etiquette. | ▪ Basics of Computer Organisation<br>• Functional Components of a Computer<br>• Input & Output Unit<br>• Memory and memory units (bit, byte, KB, MB, GB, TB)<br>• Software & Hardware<br>▪ Introduction to ICT Environment<br>• Introduction to File & Folders<br>▪ Do's and Don'ts while working on Computer | ▪ Working Computer System (Preferably Desktop)<br>▪ Class activities like Skit presentation, poster making, Quizzes, etc. | 3 | 3 | 6 |
| 2 | Data Representation and Processing-01 (Spreadsheet) | Student will be able to develop critical thinking and learn about data security & data analysis. | ▪ To know concepts of Data, Data Types, Data Security and Spreadsheets | ▪ Introduction to Data and its Type (images, numbers, text, audio, and video).<br>▪ Need for Data Storage and Data Security<br>▪ Introduction to Spreadsheet<br>▪ Creation of a spreadsheet & Entering data<br>▪ Insertion & Deletion of rows & Columns<br>▪ Use of min, max, sum, count, and average function | Open Office software suite | 3 | 6 | 9 |
| 3 | Data Representation and Processing-02 (Word Processing) | Students will be able to develop creativity and Imagination (by activities like digital poster making) | ▪ To learn the formatting in a word processing document to represent data in a well-formatted manner (letter, notice, poster) and printing it | ▪ Typing and saving text in a word processing tool.<br>▪ Formatting a word document<br>• tabs and alignment<br>• highlight<br>• font and font size<br>• Bold, Italics, Underline<br>• Changing Font Type, Size, and Colour<br>▪ Inserting Hyperlinks & Images<br>▪ Find and Replace<br>▪ print a document. | Open Office software suite | 3 | 6 | 9 |

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| 4 | Basics of Internet | Students will be able to know about the Ethical & Security issues for safe Internet Usage | ▪ To Learn the basics of the Internet, Browsers, and safely accessing the internet | ▪ Introduction to network and Internet<br>▪ World Wide Web (WWW)<br>▪ Concept of web page and website<br>▪ Internet Browsers<br>▪ Search Engine<br>▪ URL and Browsing the Net<br>▪ Safe Search, Browser Settings (add-ons, plug-ins, cookies, private browsing, etc), Digital etiquette | Firefox | 3 | 3 | 6 |
| 5 | Learn to Code-01 | Students will be able to develop basic computational thinking by Reasoning & Problem-solving. | ▪ To understand the basics of algorithm design and create scratch-based programs | ▪ Introduction to Problem Solving<br>▪ Identify different steps involved in a problem<br>▪ Introduction to Algorithms and Flow Chart<br>▪ Designing flowcharts<br>▪ Introduction to Program, Programming & Programming Language<br>▪ Introduction to Scratch and its interface<br>▪ Concept of variables.<br>▪ Concept of sequence control<br>▪ Drawing a line, Triangle, Rectangle, and Square using Scratch | Programming: Scratch Tool<br><br>Flow Chart: Google Drawing | 4 | 8 | 12 |
| 6 | Being Future Ready-01 | Students will be able to develop communication, collaboration, and research skills by using immersive experience technologies. | ▪ To understand the concept of immersive experience and use it in education | ▪ Introduction to Virtual Reality & Augmented Reality | **Virtual Reality:** Google Art and culture<br><br>**Augmented Reality:** NCERT ePathshala AR | 2 | 2 | 4 |
| 7 | Exhibition & Evaluation of Portfolio | - | - | - | - | 0 | 4 | 4 |

# Suggested Tools/Resources

| Theme | Suggested Resources |
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| **Introduction to ICT Environment** | Activities using GCompris<br>**Link : (Android) : https://play.google.com/store/apps/details?id=net.gcompris.full&hl=en**<br>**Link : (Windows) : https://gcompris.net/index-en.html**<br>**Additional Resources**<br>1. ICT in Education Class 6th (SCERT Delhi)<br>**Link :https://www.youtube.com/watch?v=n9sIGl6gBEo&feature=youtu.be**<br>2. NPTEL : ICT Basics<br>**Link : https://www.youtube.com/watch?v=KpyUBIBojIY**<br>CWSN resources: NVDA ( Non visual desktop access)<br>**Link: https://www.nvaccess.org/download/** |
| **Data Representation and Processing-01 (Spreadsheet)** | MS Excel Android App (in-app purchases)<br>**Link :https://play.google.com/store/apps/details?id=com.microsoft.office.excel&hl=en**<br>Google sheet (Web Version)<br>**Link :https://docs.google.com/spreadsheets/u/0/**<br>Open Office software suite<br>**Link :https://www.openoffice.org/**<br>Excel video training<br>**Link :https://support.microsoft.com/en-us/office/excel-video-training-9bc05390-e94c-46af-a5b3-d7c22f6990bb**<br>**CWSN (Others): use of inbuilt Accessibility tools like text to speech, alt text etc.**<br>**Google sheets :** Tools menu (spell check, personal dictionary, auto complete), Accessibility (screen reader support-for Braille support, Screen Magnifier) - FREE |
| **Data Representation and Processing-02 (Word** Processing**)** | MS Word (in-app purchases)<br>**Link :https://play.google.com/store/apps/details?id=com.microsoft.office.word** |

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| | Google Docs(Web Version)<br>**Link : https://docs.google.com/document/u/0/**<br><br>Open Office software suite<br>**Link :https://www.openoffice.org/**<br><br>Word for Windows training<br>**Link :https://support.microsoft.com/en-us/office/word-for-windows-training-7bcd85e6-2c3d-4c3c-a2a5-5ed8847eae73**<br><br>**CWSN :use of inbuilt Accessibility tools like text to speech, alt text etc**<br>**Google docs :** Tools ( spelling and grammar, dictionary, voice typing), Accessibility (screen reader support-for Braille support, Screen Magnifier) - FREE |
| **Basics of Internet** | DuckDuckGO<br>**Link: Privacy Search Engine : https://duckduckgo.com/**<br><br>Google Chrome (Android)<br>**Link :https://play.google.com/store/apps/details?id=com.android.chrome**<br><br>Google Chrome (window)<br>**Link :https://www.google.com/chrome/**<br><br>Kiddle : Safe visual search engine for kids<br>**Link: https://www.kiddle.co/**<br><br>CBSE Cyber Safety Manual<br>**Link :https://cbseacademic.nic.in/web_material/Manuals/Cyber_Safety_Manual.pdf**<br><br>**CWSN :**Web Access Tools for the Visually Impaired :- Screen Readers: A screen reader is a web accessibility software tool which reads text on the screen with a speech synthesizer or (less frequently) translates it into Braille., Screen Magnifier,<br><br>Google chrome extensions :- OpenDyslexic for Chrome<br>**Link: https://chrome.google.com/webstore/detail/opendyslexic-for-chrome/cdnapg-fjopgaggbmfgbiinmmbdcglnam?hl=en**<br><br>Text to Speech and Read aloud<br>**Link: https://chrome.google.com/webstore/detail/read-aloud-a-text-to-spee/hdhina-didafjejdhmfkjgnolgimiaplp?hl=en** |

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| **Learn to Code-01** | ScratchJr (or any other Similar App)<br>**Link :https://play.google.com/store/apps/details?id=org.scratchjr.android**<br><br>Scratch Tool<br>**Link :https://scratch.mit.edu/download**<br><br>Google Drawing<br>**Link :https://docs.google.com/drawings/**<br><br>AutoDraw<br>**Link :https://www.autodraw.com/**<br><br>Scratch Tutorials<br>**Link : https://www.youtube.com/watch?v=-3oCdNIeU_8**<br><br>**CWSN(other):** Code Jumper (A Coding Language for the Blind and Visually Impaired \| Science Times)<br>**Link : https://www.sciencetimes.com/articles/20771/20190425/a-coding-language-for-the-blind-and-visually-impaired.htm** |
| **Being Future Ready-01** | Merge Explorer<br>**Link :https://play.google.com/store/apps/details?id=com.MergeCube.EDUExplorer**<br><br>Sky View<br>**Link :https://play.google.com/store/apps/details?id=com.t11.skyviewfree**<br><br>Google Art and culture<br>(or any other Similar Tool)<br>**Link :https://artsandculture.google.com/**<br><br>CBSE Student Handbook - Level-1<br>**Link :https://cbseacademic.nic.in/web_material/Curriculum23/publication/ARVR_Student_Handbook_Level1.pdf**<br><br>CBSE Student Handbook - Level-2<br>**Link :https://cbseacademic.nic.in/web_material/Curriculum23/publication/ARVR_Student_Handbook_Level2.pdf**<br><br>**CWSN resources**: **Google Earth VR - https://arvr.google.com/earth/ - FREE WEBSITE FOR EXPERIENCE** |

## Suggested Projects:

**1. Digital Poster Making :**

This project aims to make students gather information about any issue like Climate Change, Diseases, or any other social issues using the internet and represent the cause and effects of the issue on a Digital Poster (using word processing document by embedding images and text)

For example, students can gather information about Global Warming and represent the causes and demerits on a Digital Poster using suitable Images and Text.

**2. Creating an immersive experience**

This project aims to make students explore the effectiveness of learning through the creation of an AR experience. Students can use an app like ArLoopa or 3D Bear or Devar or any other suitable app and create an immersive learning experience. For example, they can place a virtual solar system in the real world and record a video while explaining it in detail (students can record themselves while standing alongside the experience and explaining)

OR

Create a simple story in scratch. Students can choose any number of Sprite characters and create a simple story. For example they can choose any two characters, a suitable background and create a story where the two characters are communicating.